

Explainable Brain Tumor MRI Classification: EfficientNetB0 with Grad-CAM-Based Misclassification Analysis

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Abstract

The Brain Tumor MRI Dataset used in this work has over 1,400 associated public notebooks, and the overwhelming majority of them converge on the same evaluation pattern: train a convolutional model, report a single test accuracy figure, and stop. That pattern leaves a question unanswered that matters specifically for any eventual clinical use of this kind of model — when the model is right, is it right for a defensible reason, and when it is wrong, is the error a localization failure (the model looking at the wrong part of the scan) or a discrimination failure (looking at the right region but unable to separate two visually similar tumor types)? Accuracy alone cannot distinguish these two failure modes, and they imply very different fixes and very different levels of trust in the model’s output.

This work addresses that gap directly rather than adding explainability as an illustrative afterthought. A four-class brain tumor MRI classifier (glioma, meningioma, pituitary tumor, no tumor) is built on a frozen EfficientNetB0 backbone with transfer learning — chosen specifically because the dataset’s 1,400 images per class is too little to train a comparably reliable CNN from scratch, making pre-trained low-level features the more data-efficient option. On top of standard per-class precision/recall/F1 reporting, Gradient-weighted Class Activation Mapping (Grad-CAM) is applied in a second, targeted pass restricted specifically to the model’s own misclassifications, grouped by confusion pair, rather than only on a handful of correct predictions chosen for illustration. This targeted design is what lets the explainability analysis actually answer the localization-versus-discrimination question above, instead of merely showing that heatmaps exist.

The model reaches 92.62% held-out test accuracy. Per-class analysis shows glioma recall (0.75) is substantially below the other three classes (≥ 0.95) — notumor and pituitary are classified with zero test errors — and 65 of 118 total test errors are glioma misclassified as meningioma. Targeted Grad-CAM inspection of misclassified cases shows attention concentrated on or near the visible lesion in most examples, supporting a discrimination failure (visual similarity between glioma and meningioma) as the primary error driver, rather than a localization failure. However, a subset of heatmaps show a highly regular, centrally-symmetric pattern consistent with Grad-CAM

gradient saturation rather than meaningful localization, so this finding is reported as favorable but not uniformly clean, with the saturation cases explicitly excluded from the trustworthiness claim rather than counted toward it.

1 Introduction

Brain tumors are among the most serious neurological disorders, and the tumor subtype materially changes treatment planning. MRI is the standard imaging modality for this task due to its soft-tissue contrast and absence of ionizing radiation.

The Brain Tumor MRI Dataset used in this work (Nickparvar, Kaggle) has over 1,400 public notebooks associated with it. The overwhelming majority report a single test accuracy figure as the final result. This is a narrow notion of model quality for a clinical-adjacent task: two models with identical accuracy can differ sharply in which errors they make and why, and only one of these failure modes is acceptable in practice. A model that occasionally confuses two visually similar tumor types is a different (and more defensible) kind of imperfect than a model that reaches its accuracy by keying on scanner artifacts or image borders — a documented shortcut-learning failure mode in medical imaging.

This work treats explainability not as a visualization add-on but as a diagnostic tool aimed directly at that question: when the model is wrong, is it looking in the right place?

2 Dataset

The dataset contains 7,023 MRI images across four balanced classes (glioma, meningioma, pituitary tumor, no tumor), pre-split into training and testing sets with no leakage between them, per the dataset’s own documentation. Each test class contains 400 images (1,600 total). Images vary in native resolution and are resized to 160×160 during loading. Figure 1 shows representative training images across all four classes, confirming label-image consistency before training.

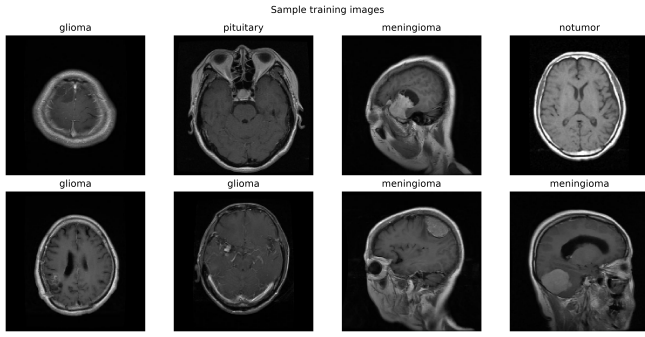


Figure 1: Representative training images across the four classes, sampled directly from the training pipeline.

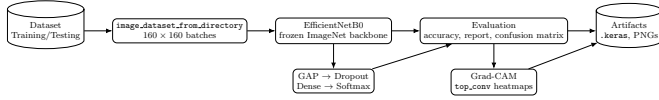


Figure 2: Pipeline: local file-system data loading, frozen-backbone transfer learning, evaluation, and post-hoc explainability. No external service, database, or experiment-tracking layer is used; this is a single-notebook research pipeline, not a production system.

3 Method

3.1 Architecture

The classifier uses EfficientNetB0 pretrained on ImageNet as a frozen feature extractor (`include_top=False`, `input_shape=(160,160,3)`), with a lightweight classification head: global average pooling, dropout (rate 0.3), a 128-unit ReLU dense layer, and a 4-unit softmax output. Of 4,214,055 total parameters, only 164,484 (the head) are trainable.

Transfer learning with a frozen backbone was chosen deliberately over training a CNN from scratch. With 1,400 training images per class, a from-scratch CNN has insufficient data to learn generalizable low-level features and tends toward over- or under-fitting on dataset-specific noise. EfficientNetB0’s ImageNet-pretrained early layers already encode general low-level visual structure (edges, textures, gradients) that transfers reasonably well, allowing the available data to be spent entirely on learning the task-specific decision boundary rather than re-deriving generic visual features.

3.2 Training Procedure

The training directory is split 85%/15% into training and validation subsets (seed fixed at 42 for reproducibility); the testing directory is loaded separately with `shuffle=False` so that prediction order can later be aligned with ground-truth labels for misclassification analysis. Training uses categorical cross-entropy loss and Adam (initial learning

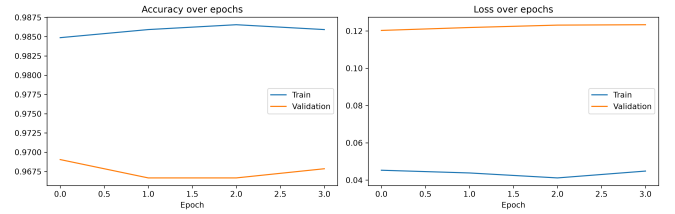


Figure 3: Training and validation accuracy/loss across the four completed epochs.

rate 10^{-3}). Early stopping (patience 3, monitoring validation loss, restoring best weights) and `ReduceLROnPlateau` (factor 0.5, patience 2, minimum learning rate 10^{-6}) control training length and step size. The recorded run reached a training accuracy of 98.6% and validation accuracy of 96.8% at termination (Figure 3), with the learning rate reduced to 1.56×10^{-5} by the final logged epoch — training ended due to validation-driven scheduling rather than a fixed epoch budget. Training accuracy and loss are stable rather than diverging from validation, indicating no severe overfitting over the four logged epochs, though the small absolute gap between training and validation loss is consistent with the limited number of epochs completed before early stopping.

3.3 Grad-CAM

Grad-CAM is computed from the gradient of the predicted class score with respect to the activations of the final convolutional layer (`top_conv`). These gradients are spatially averaged into per-channel importance weights, applied to the corresponding feature maps, passed through a ReLU, normalized to $[0, 1]$, resized to the input resolution, and alpha-blended over the original image using OpenCV’s JET colormap.

Critically, Grad-CAM is applied in two passes in this work: a general pass over a sample of test images, and a second, targeted pass restricted to images the model misclassified, grouped by the most frequent true→predicted confusion pairs. The second pass is what allows the explainability analysis to speak directly to model trustworthiness, rather than serving only as illustration on cases the model already got right.

4 Results

4.1 Classification Performance

The model achieves 92.62% accuracy and 0.4220 loss on the 1,600-image held-out test set. Table 1 reports per-class precision, recall, and F1.

The per-class breakdown is the first finding that an aggregate accuracy figure conceals: glioma recall (0.75) is

Table 1: Held-out test classification report (400 images per class).

Class	Precision	Recall	F1	Support
Glioma	0.96	0.75	0.85	400
Meningioma	0.85	0.95	0.90	400
Notumor	0.94	1.00	0.97	400
Pituitary	0.96	1.00	0.98	400
Macro avg.	0.93	0.93	0.92	1600

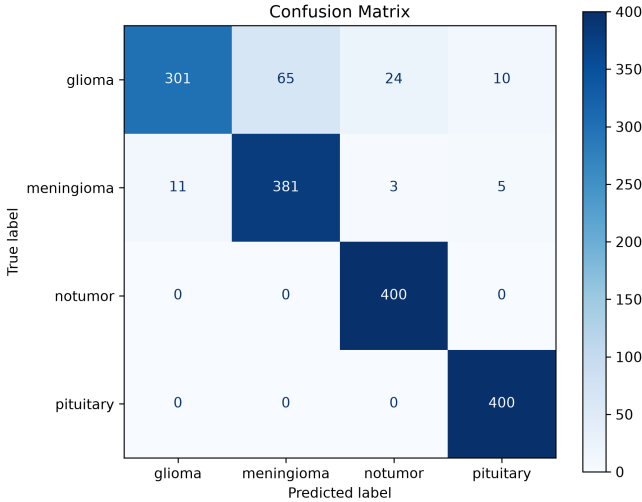


Figure 4: Confusion matrix on the 1,600-image held-out test set. Notumor and pituitary are classified with zero errors; nearly all confusion occurs between glioma and meningioma.

markedly lower than the other three classes, meaning one in four glioma scans is misclassified, despite glioma precision being the highest of any class (0.96). This asymmetry — high precision, low recall — indicates the model is conservative about predicting glioma (rarely wrong *when* it predicts glioma) but frequently fails to predict glioma *when it should*, with those cases absorbed into adjacent classes.

4.2 Misclassification Distribution

118 of 1,600 test images were misclassified (7.4% error rate). Figure 4 shows the full confusion matrix: notumor and pituitary are classified perfectly (400/400 each), while essentially all error mass falls on glioma and meningioma.

Glioma→meningioma alone accounts for 65 of 118 errors (55%). Glioma→notumor (24 cases) is smaller but more clinically severe, since it represents a true tumor predicted as a healthy scan — a false negative in the highest-stakes direction for this task.

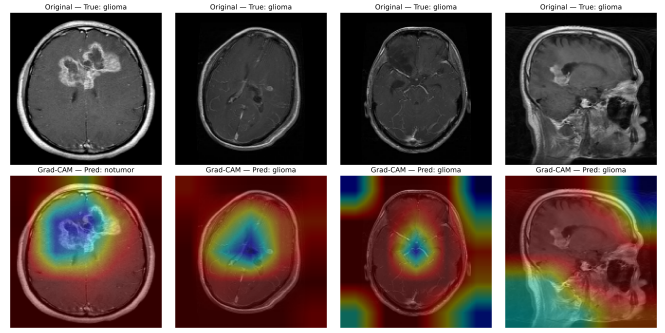


Figure 5: Grad-CAM on a general sample of test images. The first panel’s heatmap aligns closely with the visible lesion. The second and third panels show a sharply centered, symmetric diamond pattern unrelated to visible anatomy — most likely a gradient-saturation artifact rather than genuine localization (see Section 5). The fourth panel’s heatmap falls mostly outside brain tissue, on lower facial/neck structure.

4.3 Grad-CAM Evidence

Figure 5 shows Grad-CAM applied to a general sample of test images (a mix of correct and incorrect predictions, not specifically targeted at errors). Figure 6 shows the targeted pass on confirmed misclassifications for the meningioma→glioma confusion pair — the only confusion pair retained as a saved figure due to the file-overwrite issue noted in Section 5.

The evidence is more mixed than a single clean narrative. In the targeted misclassification figure (Figure 6), two of three examples show attention reasonably close to the visible lesion despite the wrong predicted label, consistent with the project’s central hypothesis: glioma and meningioma can be visually similar enough that even attention placed on the correct structure does not guarantee correct discrimination. However, the center example in the same figure, and two of four examples in the general sample (Figure 5), show a highly regular, centrally-symmetric heatmap shape that does not track any visible anatomical feature. Because this exact diamond/V-shaped pattern recurs across unrelated images, it is more consistent with Grad-CAM gradient saturation at high prediction confidence than with a meaningful claim about where the model is or is not looking, and should not be interpreted as evidence of either correct or incorrect localization.

Only one confusion pair (meningioma→glioma) was recoverable as a saved figure for this run, due to the implementation issue described in Section 5; the glioma→meningioma and glioma→notumor pairs — which together account for 89 of 118 errors — were generated and visually inspected during execution but were not preserved to disk. The qualitative impression from that inline inspection was consistent with the pattern above: attention mostly on or near visible tumor tissue, with occasional saturation

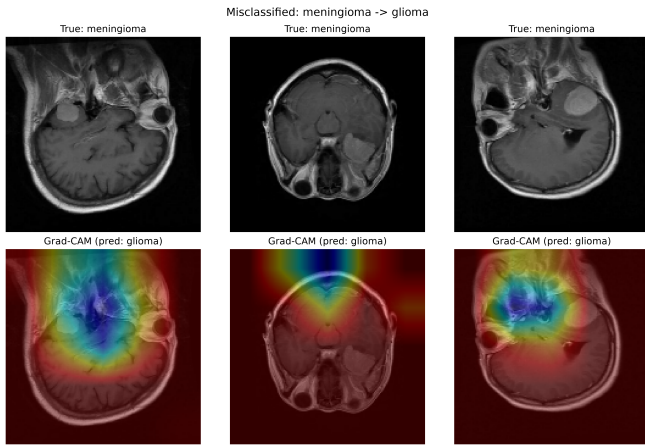


Figure 6: Targeted Grad-CAM on confirmed meningioma→glioma misclassifications. Left and right panels show attention near the visible lesion; the center panel shows a symmetric V-shaped pattern centered at the superior skull margin, not aligned with the visible mass.

artifacts. A re-run with the filename fix applied would allow this claim to be supported with retained figures for all three top confusion pairs rather than one.

5 Limitations

The backbone remains frozen throughout training; fine-tuning upper EfficientNet blocks may improve glioma/meningioma discrimination specifically but was not evaluated here. Grad-CAM interpretation in this work is qualitative — heatmaps are visually inspected rather than scored against ground-truth tumor segmentation masks, which this dataset does not provide. A quantitative attention-overlap metric (e.g., thresholded heatmap area inside a brain-tissue mask versus border regions, aggregated across the full test set) would convert the qualitative claim in Section 4.3 into a measurable one and is the most direct next step.

A second, implementation-level limitation was identified directly in the misclassification-visualization cell: the figure-saving call sits inside the per-confusion-pair loop and writes to a fixed filename, so each iteration overwrites the previous confusion pair’s saved figure on disk. As a result, only the last-processed confusion pair (meningioma→glioma) was retained as a saved file, even though all three top confusion pairs were correctly computed and displayed inline during execution. The fix is to include the confusion-pair label in the output filename so each figure is preserved separately; this does not affect the correctness of the underlying analysis, only which figures survive to disk.

A further observation from the retained Grad-CAM figures is that some heatmaps exhibit a highly regular, centrally-

symmetric pattern (e.g., a sharply centered diamond or V-shape) that does not track visible anatomical structure. This is consistent with gradient saturation at high prediction confidence, a known limitation of vanilla Grad-CAM, rather than a meaningful localization signal, and should be treated as inconclusive evidence rather than counted toward either a positive or negative trustworthiness finding. Finally, the workflow is notebook-centric with no automated testing, experiment tracking, or environment pinning beyond the printed TensorFlow/GPU configuration at runtime.

6 Conclusion

This work combines EfficientNetB0 transfer learning with targeted Grad-CAM analysis of the model’s own errors to ask a question that accuracy alone cannot answer: when this model is wrong, is it looking at the right thing? The evidence leans toward yes — most inspected misclassifications show attention on or near the visible lesion, consistent with a glioma/meningioma discrimination problem driven by genuine visual similarity rather than reliance on irrelevant image regions — but the picture is not uniformly clean: a recurring, anatomically-uncorrelated heatmap pattern consistent with gradient saturation appears in a minority of cases and limits how strongly the localization claim can be stated. This reframes the project’s contribution: not a new architecture, but evidence, with explicit caveats, bearing on whether an architecture’s accuracy is trustworthy — a more honest and more relevant answer than either an unqualified confirmation or an unexamined accuracy figure would provide.